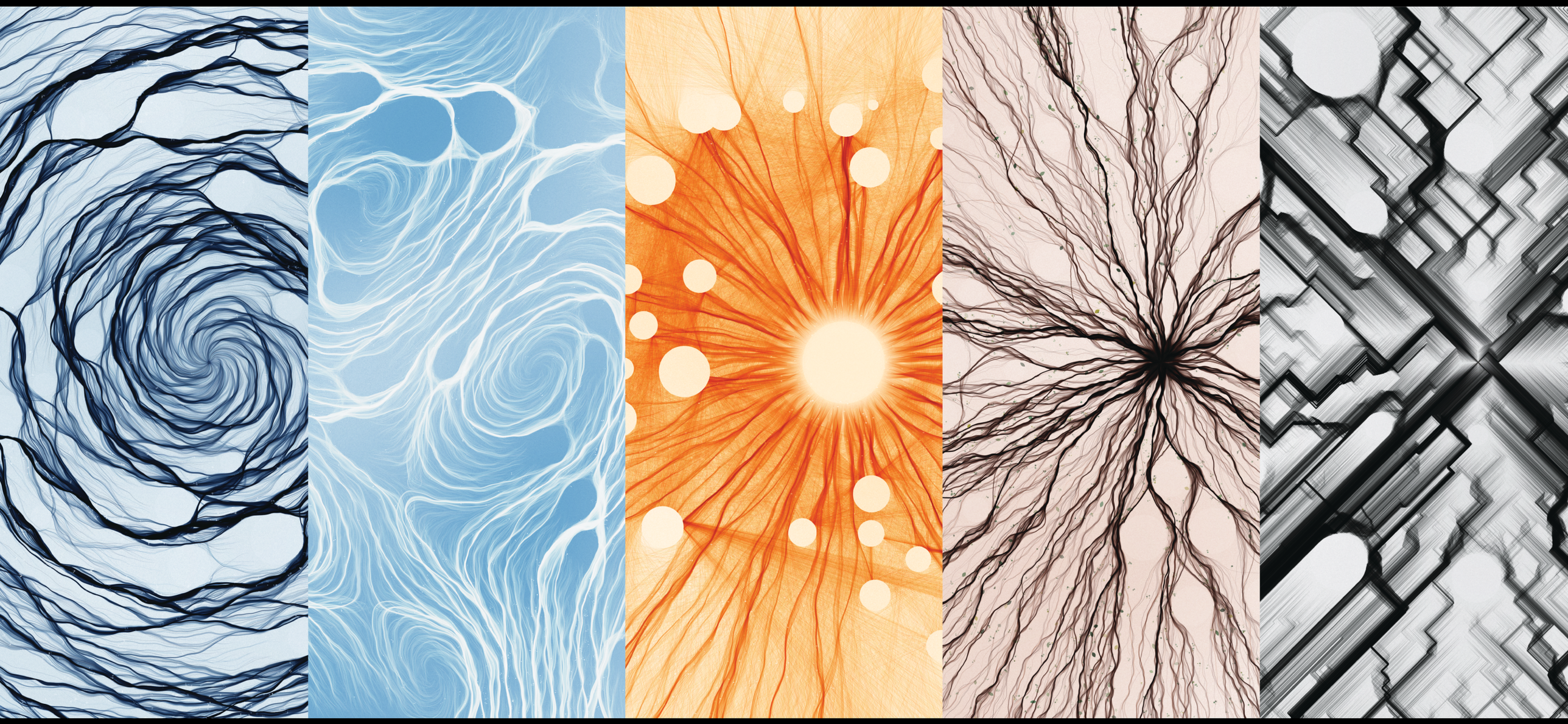
JMM331/622



INFOGRAPHICS AND DATA VISUALIZATION

Alberto Cairo • University of Miami • www.thefunctionalart.com • Twitter: @albertocairo

To improve attention in class:

1. Turn your camera on.
2. Disable message alerts in your computer.
3. Close your web browser and social media applications.
4. Put your cell phone away where you can't reach it easily.

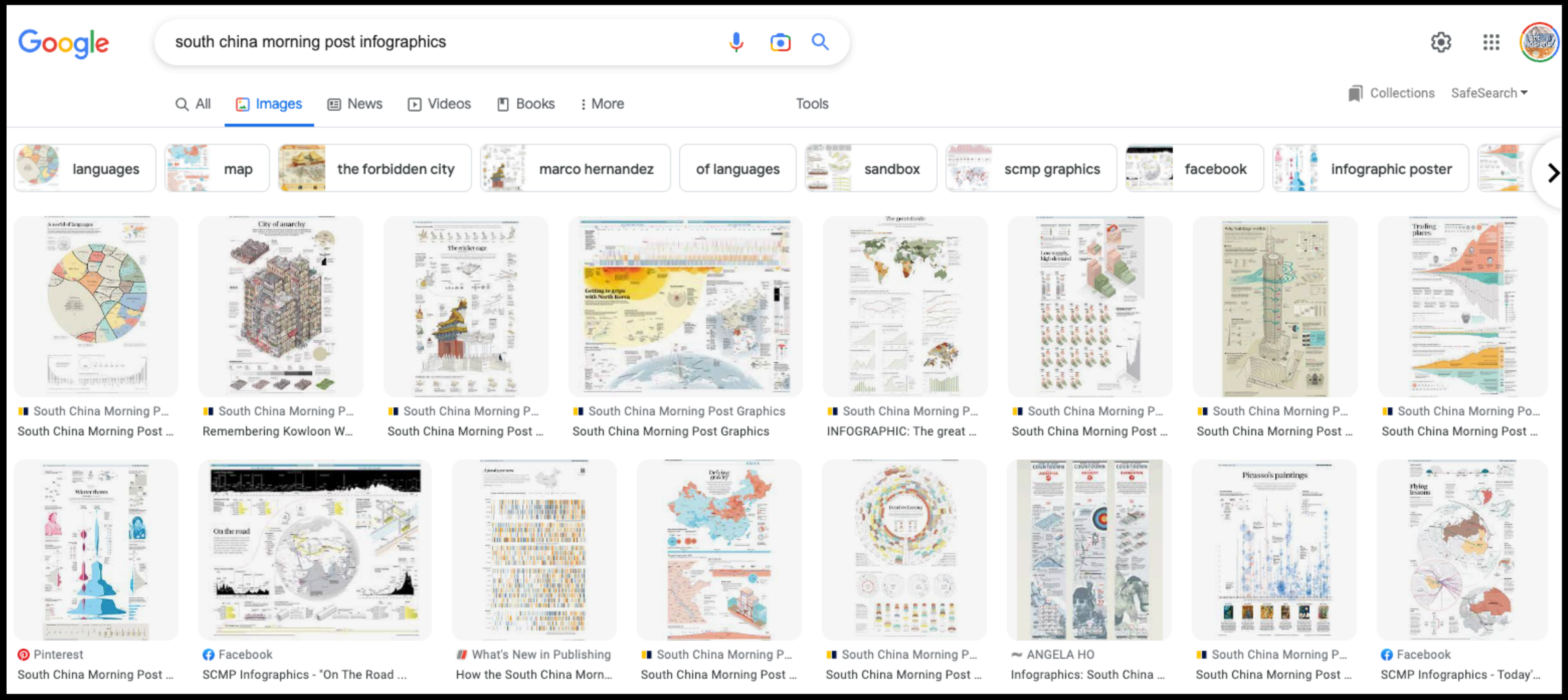
Feel free to unmute yourselves and ask questions whenever you want.

Recommendations

South China Morning Post Infographics

Google search: https://tinyurl.com/ystyv5ha
Their website: https://www.scmp.com/infographic/

(They use a similar workflow to the one we're using in this class)

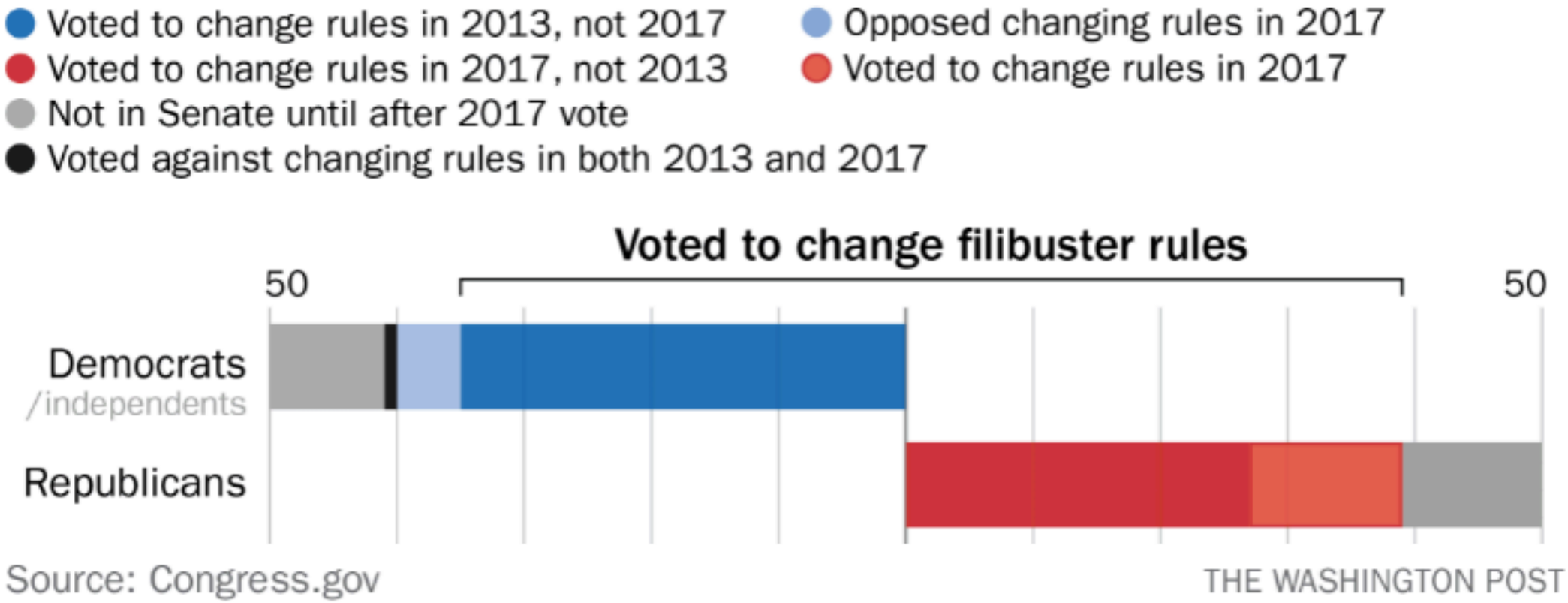


Recommendations

How to Read this Chart newsletter

https://www.washingtonpost.com/newsletters/how-to-read-this-chart/

Is this an overcomplicated way to show this data? Sure! **I'm not vain. I can handle complaints**. So in that same article, I offered another way of looking at the same thing.



New version of the Syllabus uploaded

IMPORTANT:
NO CLASS ON THURSDAY, September 7th

Exercises from last week…

HOMEWORK FOR THE NEXT WEEK AND A HALF:

Due Friday, September 15th before the end of the day

Housekeeping:

Don't forget your readings! 2-3 chapters or articles per week.

Tutorials

Illustrator Lesson 4: Charts and maps

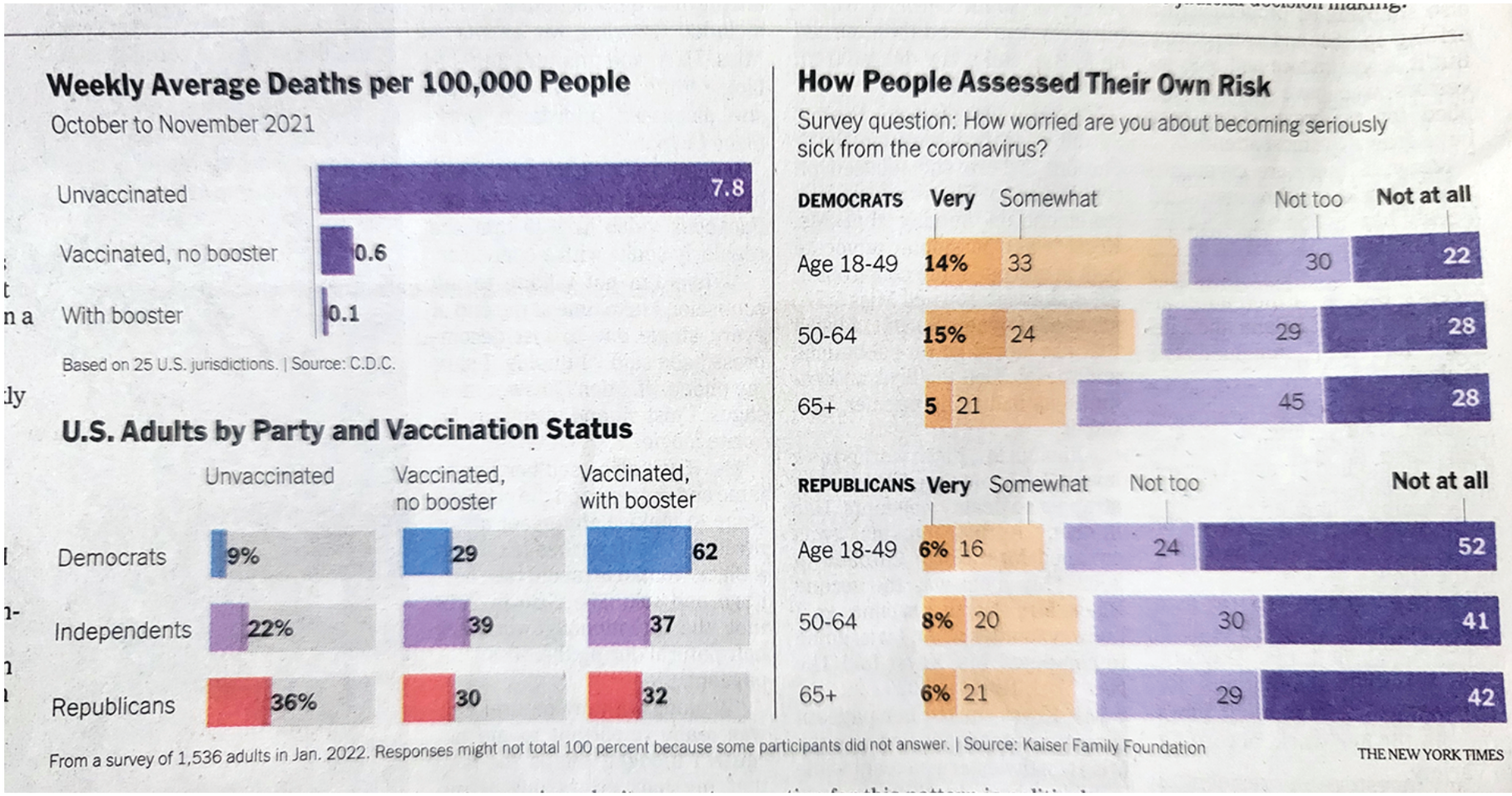
RawGraphs tutorials:
https://rawgraphs.io/learning

Flourish tutorials:
https://help.flourish.studio/

Assignment:

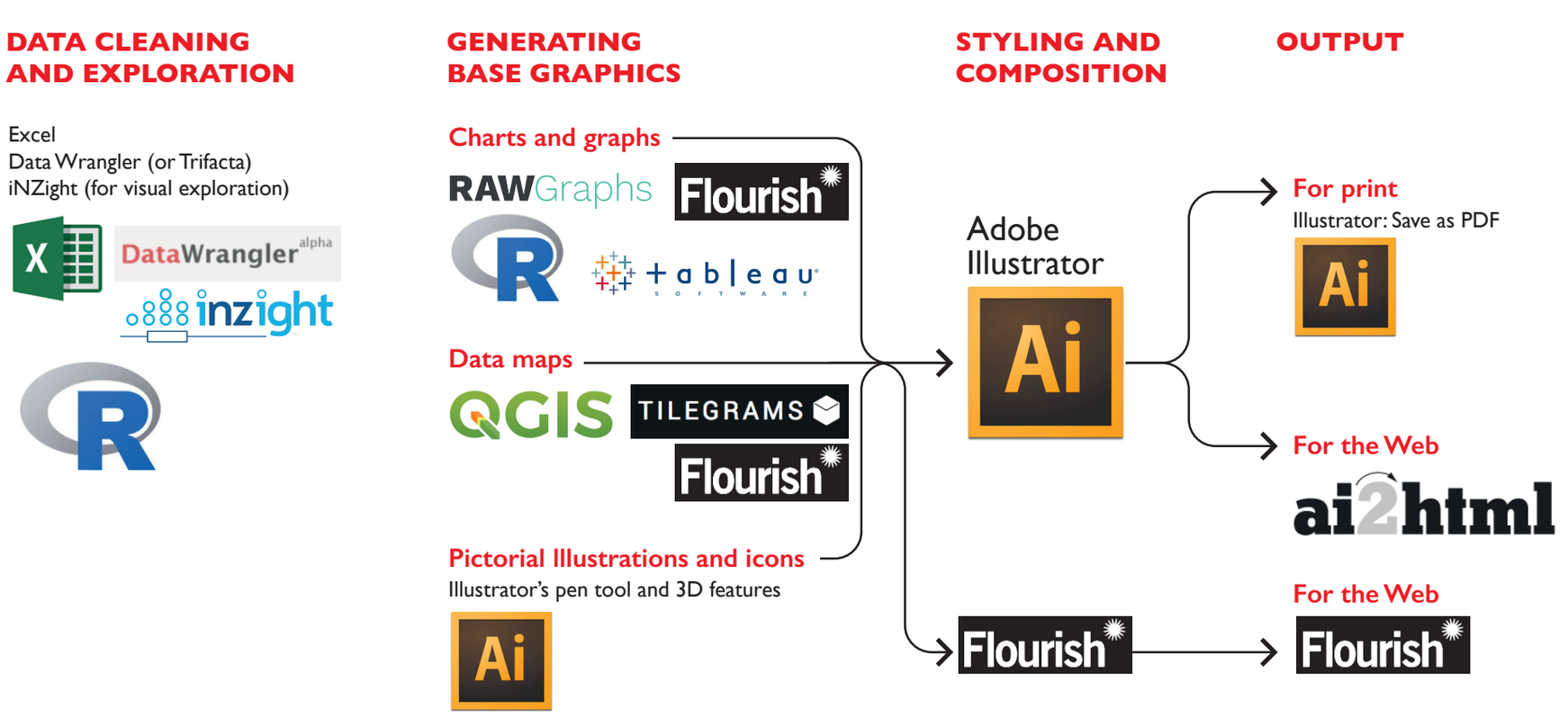
Design an interesting infographic made of AT LEAST FOUR maps and/or graphs.

Simple charts like these can be designed directly in Adobe Illustrator.



But more often than not you'll follow a workflow like this, generating your charts in other programs, and then importing them into Illustrator for styling, customization, and layout.

MY WORKFLOW

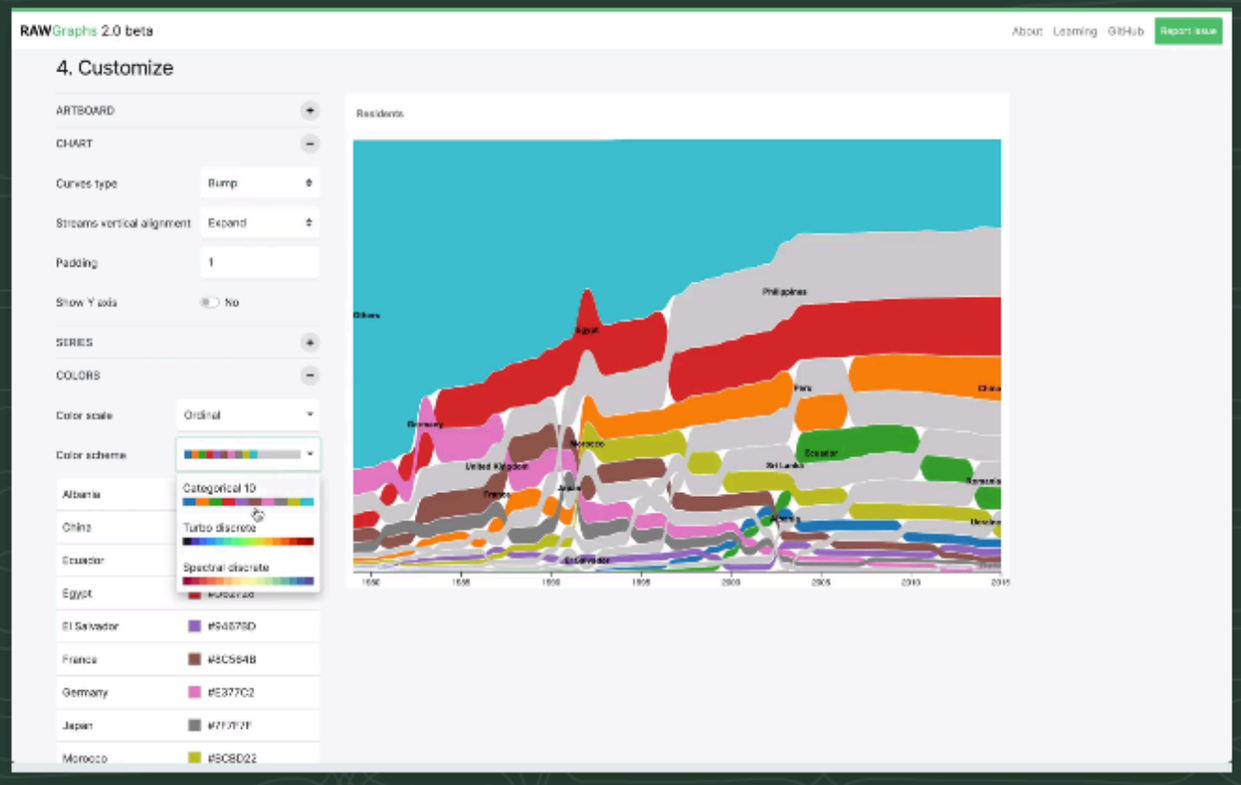


RAWGraphs

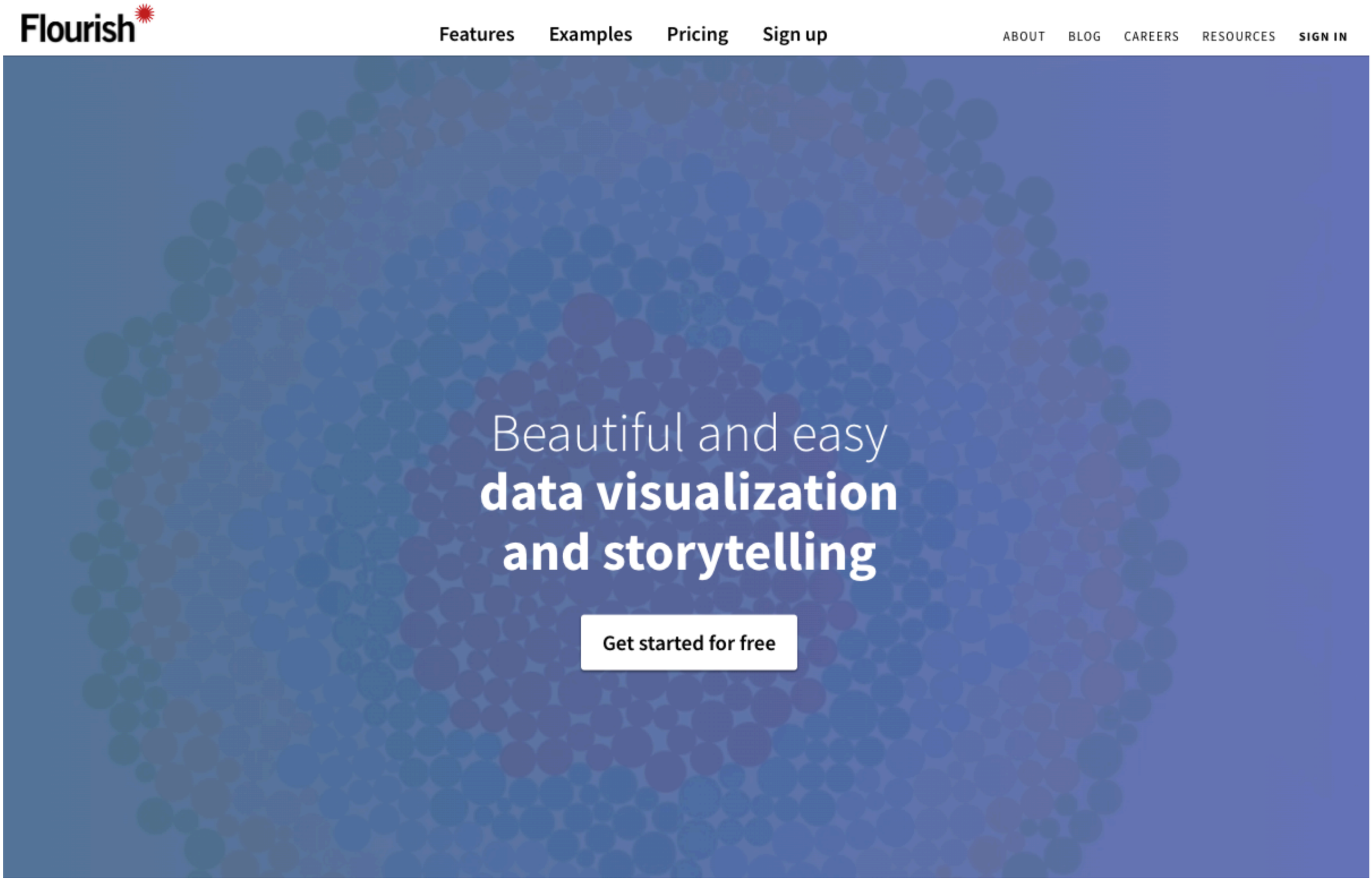
About News Support us Learning Resources **Use it now!**

A free and open source tool for data visualization

Use it now! **Github**



https://www.rawgraphs.io/



http://flourish.studio/

Assignment due on Friday, September 15th at the end of the day

Size and orientation: Tabloid size
Topic: Your choice
Data: Your choice
Graphics: at least **FOUR**, and at least one of them **must be a map of some kind (see topic suggestions in following slides)**

FORMAT: PDF. No other format is acceptable

You can generate your charts/maps with either Illustrator, RawGraphs, Flourish, or any combination of them, but then you'll have to bring them together into a composition in Adobe Illustrator, and add a main title, an introduction, a title for every section, etc.

As always, turn your exercise here:

https://drive.google.com/drive/folders/1Vuu0aO8N-OREi4fxfG1TemyFKh5KAvut?usp=drive_link

(Don't forget to name your file correctly: name_lastname_infographicExercise.pdf or something similar)

Example that shows the elements you should include in your design

(notice that this story only has 2 graphics, though. Yours should have at least 4 sections/graphics)

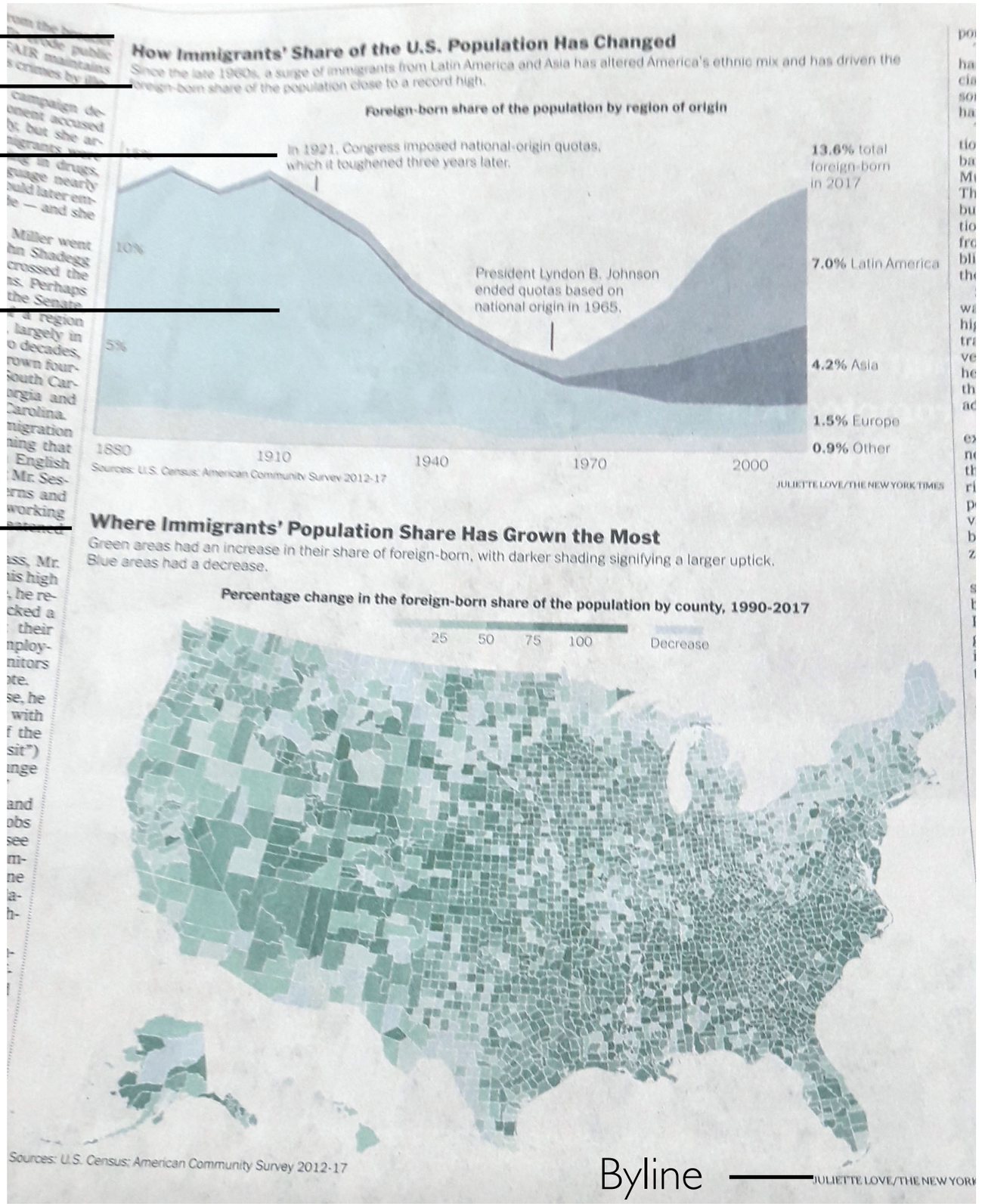
Main headline

Introduction

Annotations

Content: graphs, maps, illustrations, etc.

Titles and intros on every section



Sources

Byline

https://archive.nytimes.com/www.nytimes.com/imagepages/2011/09/04/opinion/04reich-graphic.html?_r=1

Example by a former student

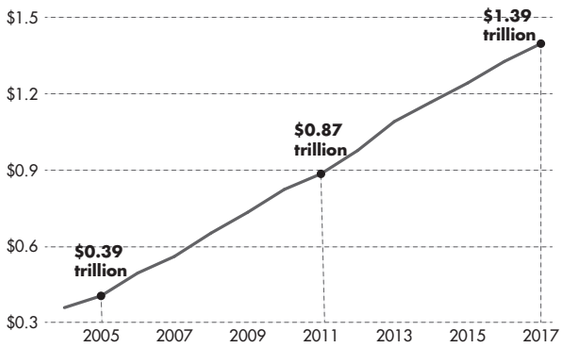
HOW **STUDENT LOAN DEBT** IS CRUSHING MILLENIALS' HOME OWNERSHIP DREAMS

by Adam Clarke

Homeownership rates among young Americans are at their lowest point in three decades. For many of these Americans in their 20s and 30s, the prospect of purchasing a home is perceived as an unreachable goal. The fast rising burden of tuition has caused student loan debt in the United States to grow rapidly, **surpassing $1.4 trillion in 2018**. The bulk of this debt is held by millenials, and it is one of the most significant factors keeping them out of the home buying market. A recent survey by the National Association of Realtors showed that more than **80% of millenials say student debt has limited their plans to purchase a home.**

WHILE STUDENT DEBT CONTINUES TO RISE...

TOTAL STUDENT LOAN DEBT (IN TRILLIONS)
FROM 2004-2017

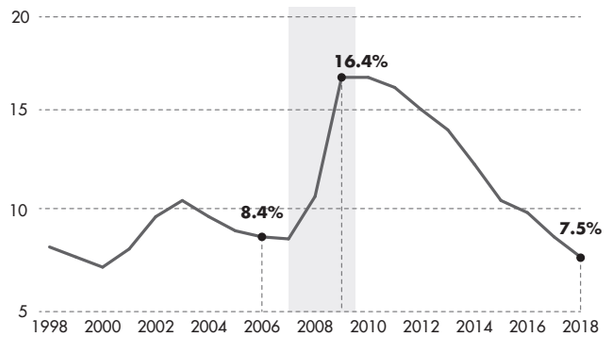


(source: Federal Reserve Bank of New York Consumer Credit Panel)

Student loan debt has grown at a remarkably fast rate, more than **tripling** from $391 billion in 2005 to $1.39 trillion in 2017. Tuition increases are the biggest reason for growth, forcing students to take on more loans to meet the cost of their tuition bills. Additionally, **government disinvestment in higher education** has made families more reliant on debt to fund college costs.

...THE UNDEREMPLOYMENT RATE REMAINS HIGH

U-6 UNDEREMPLOYMENT PERCENTAGE
FROM 1998-2018

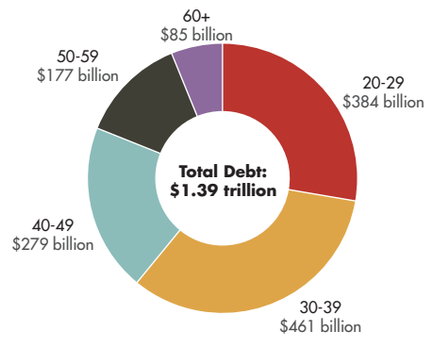


(source: Bureau of Labor Statistics)

Underemployment is often measured using the Bureau of Labor Statistics U-6 unemployment rate. This rate measures total unemployed individuals, plus marginally attached workers, plus individuals employemed part-time for economic reasons. **This rate peaked at 16.4% during the financial crisis and recession of 2007-2010.** In 2018, it reached its pre-recession levels, dropping to 7.5% in July 2018.

ADULTS 20-39 HOLD 61% OF STUDENT DEBT...

TOTAL STUDENT LOAN DEBT BY AGE GROUP
IN 2017

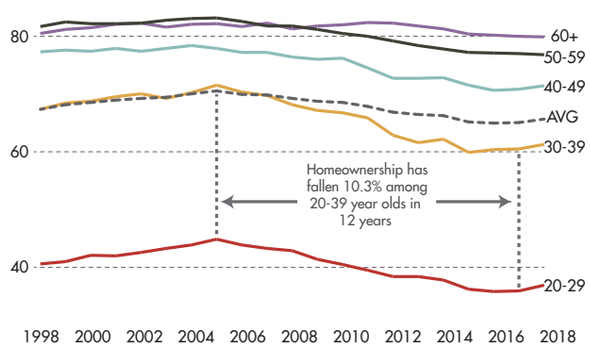


(source: Federal Reserve Bank of New York Consumer Credit Panel)

The largest holders of student debt in 2017 were adults aged 30-39. Not far behind were more recent graduates in their 20s. **Combined, these groups account for 61% of the total student loan debt in the U.S. in 2017.** Because these individuals often do not make enough money to make full payments towards their loan, and carry their debt into their 30s and 40s.

...AND ARE LEAST LIKELY TO OWN THEIR HOME

HOMEOWNERSHIP RATE BY AGE GROUP
FROM 1998-2018

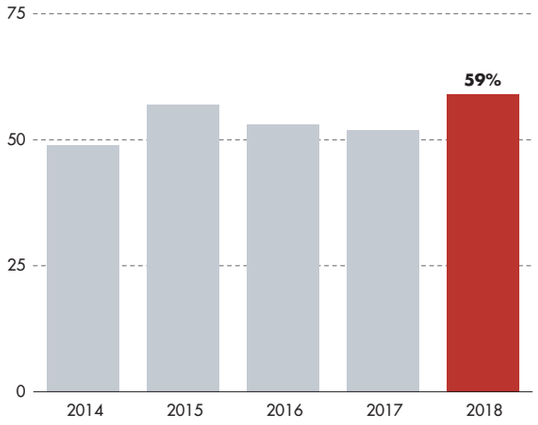


(source: U.S. Census, Survey of Consumer Finances)

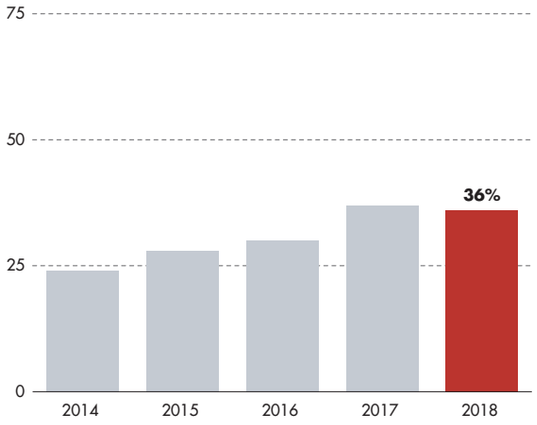
Since 2005, homeownership rates have stayed mostly steady for adults 40 and older. **During that same time, the rate for adults in their 20s and 30s has fallen more than 10%**. In 2005, 43.4% of adults ages 20-29 owned their home. That has fallen to 34.4% in 2017. In that time, the rate for adults 30-39 fell from 70.1% to 59%. While many factors have contributed to this decline, the burden of student debt has played a significant role.

YOUNG AMERICANS ARE CITING STUDENT DEBT AS AN OBSTACLE TO HOME OWNERSHIP

PERCENTAGE OF 18-24 YEAR OLDS
WHO SAY THEIR STUDENT LOAN DEBT IS
AN OBSTACLE TO BUYING A HOME



PERCENTAGE OF 18-24 YEAR OLDS
WHO KNOW SOMEONE WHO DELAYED
BUYING A HOME BECAUSE OF STUDENT LOAN DEBT



(source: NeighborWorks America at Home Survey)

THREE IMPORTANT FACTORS TO OBTAIN A HOME LOAN

Among the many factors considered by lenders when evaluating someone for a home loan, three of the most important are a person's credit score, household debt, and savings available to put towards a down payment.

MEDIAN	20-29	30-39	40-49	50-59	60+
CREDIT SCORE	631	634	655	700	730
HOUSEHOLD DEBT	$14,446	$32,698	$42,412	$42,628	$32,757
SAVINGS	$2,000	$5,000	$6,500	$12,000	$15,000

WILL IT GET BETTER ANY TIME SOON?

TOTAL STUDENT DEBT IS STILL GROWING

Nationwide, universities are increasing tuition, fees, and student housing costs. With grant money drying up and faced with little alternative, students continue to turn to student loans to pay the additional costs. Recent projections show that the total student debt may surpass $2 trillion as early as 2021.

DELINQUENCY RATES CONTINUE TO GO UP

Current delinquency rates for student loans are 22 percent, but the Brookings Institute projects that the default rate will reach 40 percent by 2024. As Americans go into bankruptcy from their student debt, the prospect of homeownership will become even more out of reach.

HOME VALUES ARE INCREASING

The financial crisis that started in 2008 had a significant effect on the prices of homes. But as the economy recovers, prices are skyrocketing. In the last 10 years, average home prices have increased as much as 60% in major metropolitan areas, where Millenials tend to cluster.

Example by a former student

You can find these projects, and many others, in the Examples folder on Blackboard. Use them as inspiration for your work.

COLOMBIA'S VIOLENT DISPLACEMENT

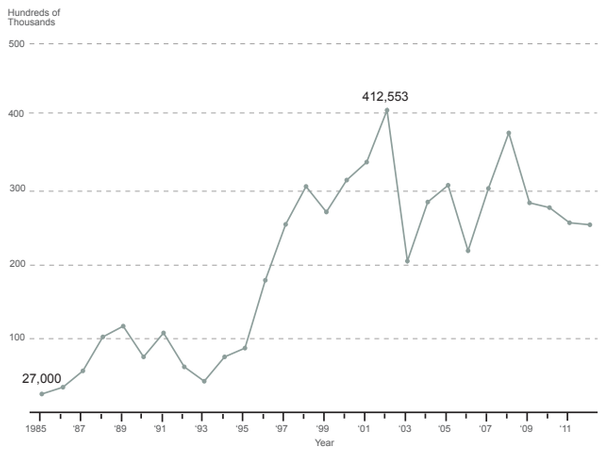
How Colombia is handling the victims of the armed conflict. Is there any hope for a better future?

Ever since the armed conflict began in Colombia more than 50 years ago, many people have become victims of the war that is now part of the daily basis. The victims are mostly from the rural zones of the country. Internally Displaced People, IDP, are people who were taken out of their lands and properties against their will, and are obligated to look for shelter, most of the times, in other regions.
The guerrilla groups FARC (Revolutionary Armed Forces of Colombia), the ELN (National Liberationist Army), paramilitary forces such as AUC (United Self-Defense Forces of Colombia), and the BACRIM (Emergent Criminal Gangs) cause the violent internal displacement in Colombia.
Once displaced by violence, people tend to stay displaced for years. This has increased the uncertainty of a better future, and it has worsened the quality of life in the whole country.

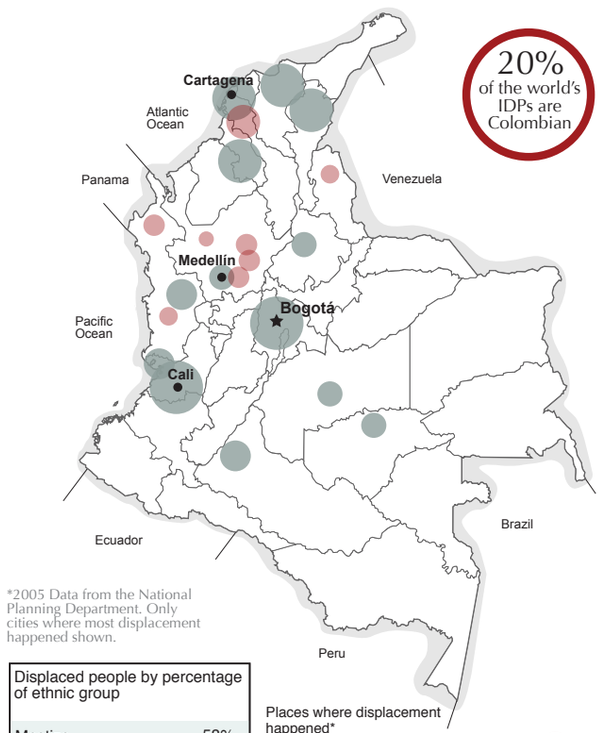
Angelica Quintero
Sources: CODHES, Univalle, UNRA, UN, El Tiempo, El País, Norwegian Refugee Council

Internal Displacement Does Not Stop

Amount of Internally Displaced People in Colombia per year...



Where do These People Go?



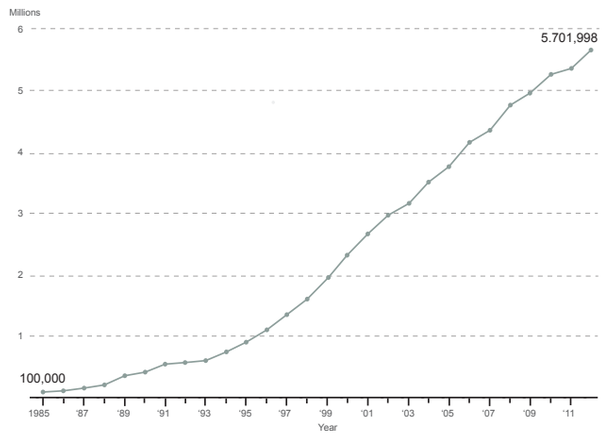
*2005 Data from the National Planning Department. Only cities where most displacement happened shown.

Displaced people by percentage of ethnic group	
Mestizo	58%
White	20
Mulatto	14
Black	4
Mixed Black/ Amerindian	3
Amerindian	1

Places where displacement happened*
5,000-10,000 10,001-15,000 15,001-25,000 More than 50,000

Places where displaced people moved*
10,000-20,000 30,000-50,000 60,000-80,000 More than 90,000

Amount of accumulated Internally Displaced People in Colombia



Surviving Displacement in an already chaotic economy

As soon as they move in to cities, Internally Displaced People from Colombia have to live in shantytowns, slums or houses where they can rent rooms per day. Many times, these houses are used as crack houses. Displaced people usually move in groups and families. Most of the times, only men can work while the women take care of the children. Many Internally Displaced Women end up begging for money at the traffic lights with their children while their husbands work as day laborers.

$5
Average daily income of an internally displaced family

$5
Average cost of renting a room per day

10%
of the people cannot supply their families with the minimum requirements of a healthy diet

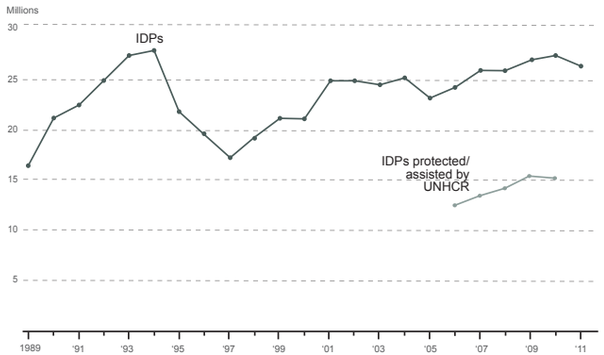
$367
Per month. Estimated Poverty Line

$299
Minimum wage per month

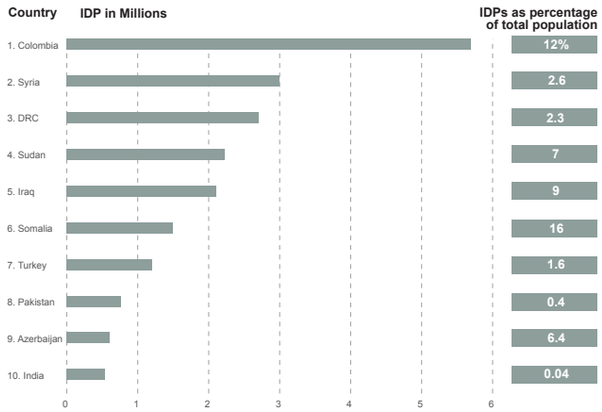
+55%
Of people earn only the minimum wage

Internal Displacement Worldwide

Internal Displacement is a global issue. The United Nation's Refugee Agency does not only help people seeking asylum in other places, but also looks to protect those who have been internally displaced and have no place to go.



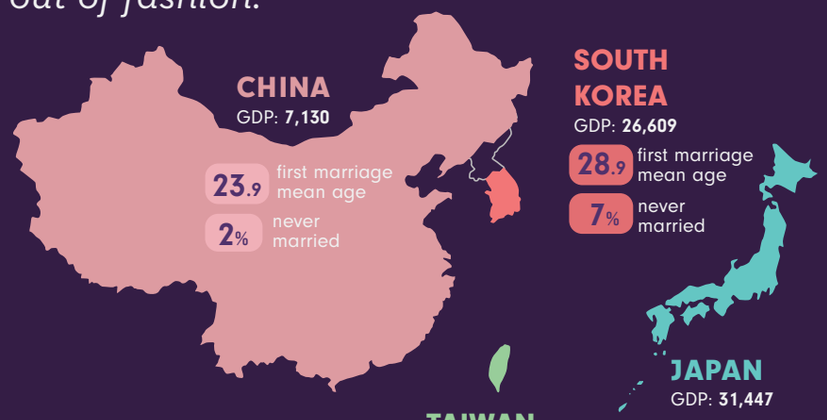
Internally Displaced People by Country



Example by a former student

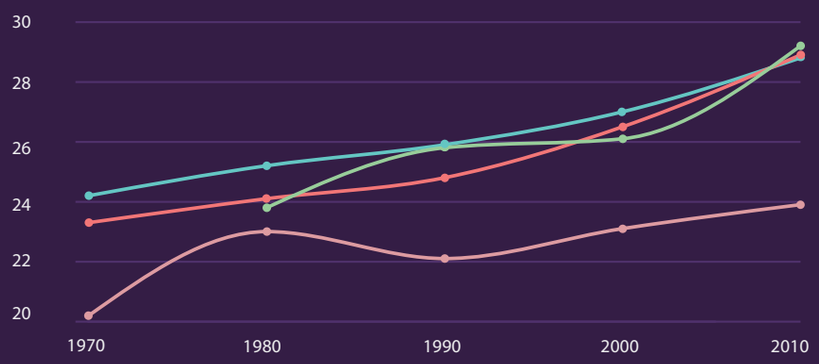
Why are East Asian women marrying later?

Love may be in the air, but commitment is out of fashion.



Over recent decades. the universality of marriage has been on a decline. The porportion of women remaining single in their 30s and 40s has climbed exponentially, especially in the large cities and among the better educated.

Mean age at first marriage, Female



East Asian family dynamics

East Asian families place high emphasis in 'strong' family ties. It is common for married couples to live together with their in-laws and children all under one roof. One particularly distinctive feature of East Asian families is a clearly-defined gender division of labor within the family.

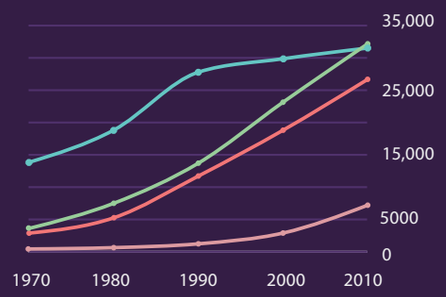
The patriarchal family system has put women at social and economic disadvantages. Many traditional marriages are characterized by the lack of independence.

Sources: Asia Research Insititute, UNESCO Institute for Statistics
By Katy Huang

Economy on a rise

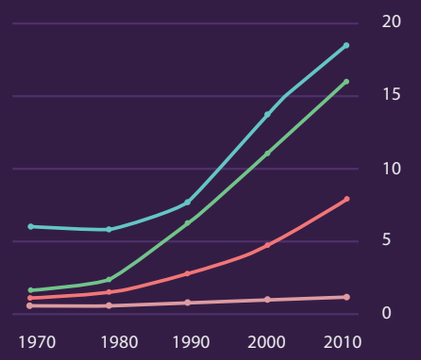
As the economy of East Asian countries improves, women are better educated, and can support themselves financially. They longer see the traditional family as the only way to lead a fulfilling life.

GDP per capita



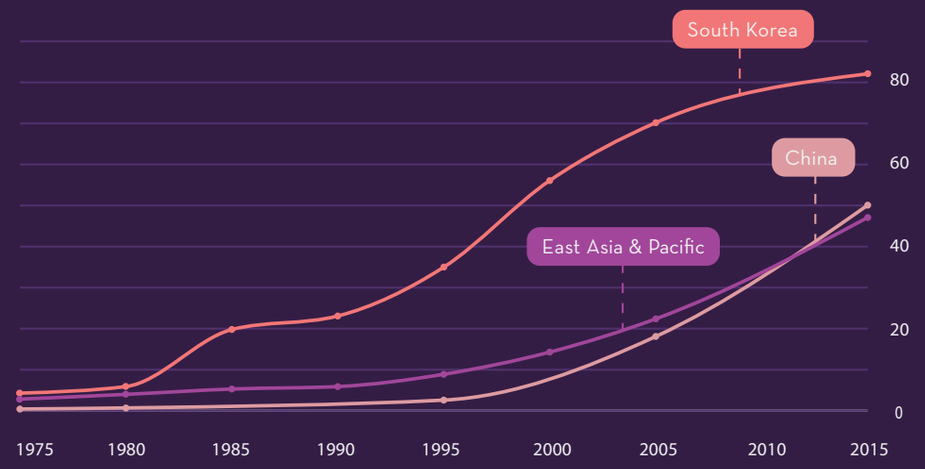
The declining of marriage

Women aged 35-49 who have never married, % of total



An educated choice

Gross enrollment ratio, Tertiary Education %, female

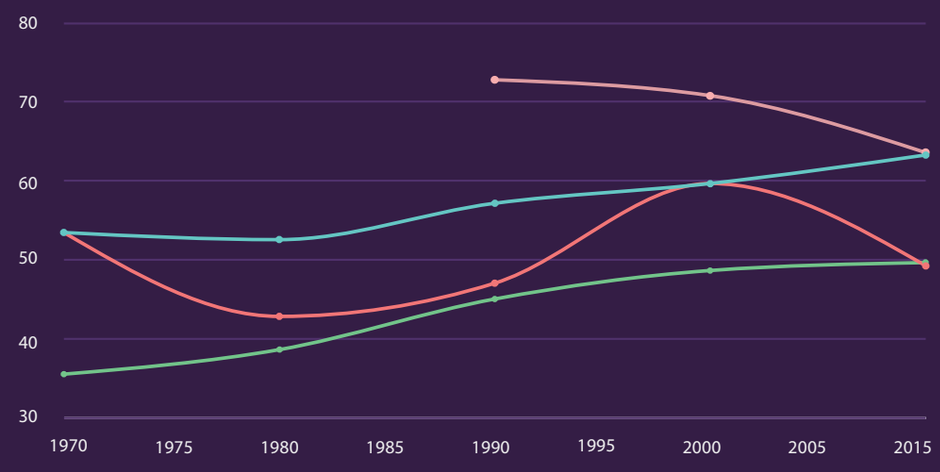


Two forces are giving women more autonomy: education and jobs. Women's education in East Asia has impacted dramatically over the past 30 years. In South Korea now, women earn half of all master's degrees. Education increases job prospects for women. Therefore allowing them to be earn more so they don't need to solely rely on their significant other.

However, in Japan, a strong tradition of women being housewives after marriage still persists. One of the reasons why women are now opting out of marriage is to pursue a career in what they love. In sum, Freedom and independence are the two factors that explains the declining rate and delayed marriage in East Asia.

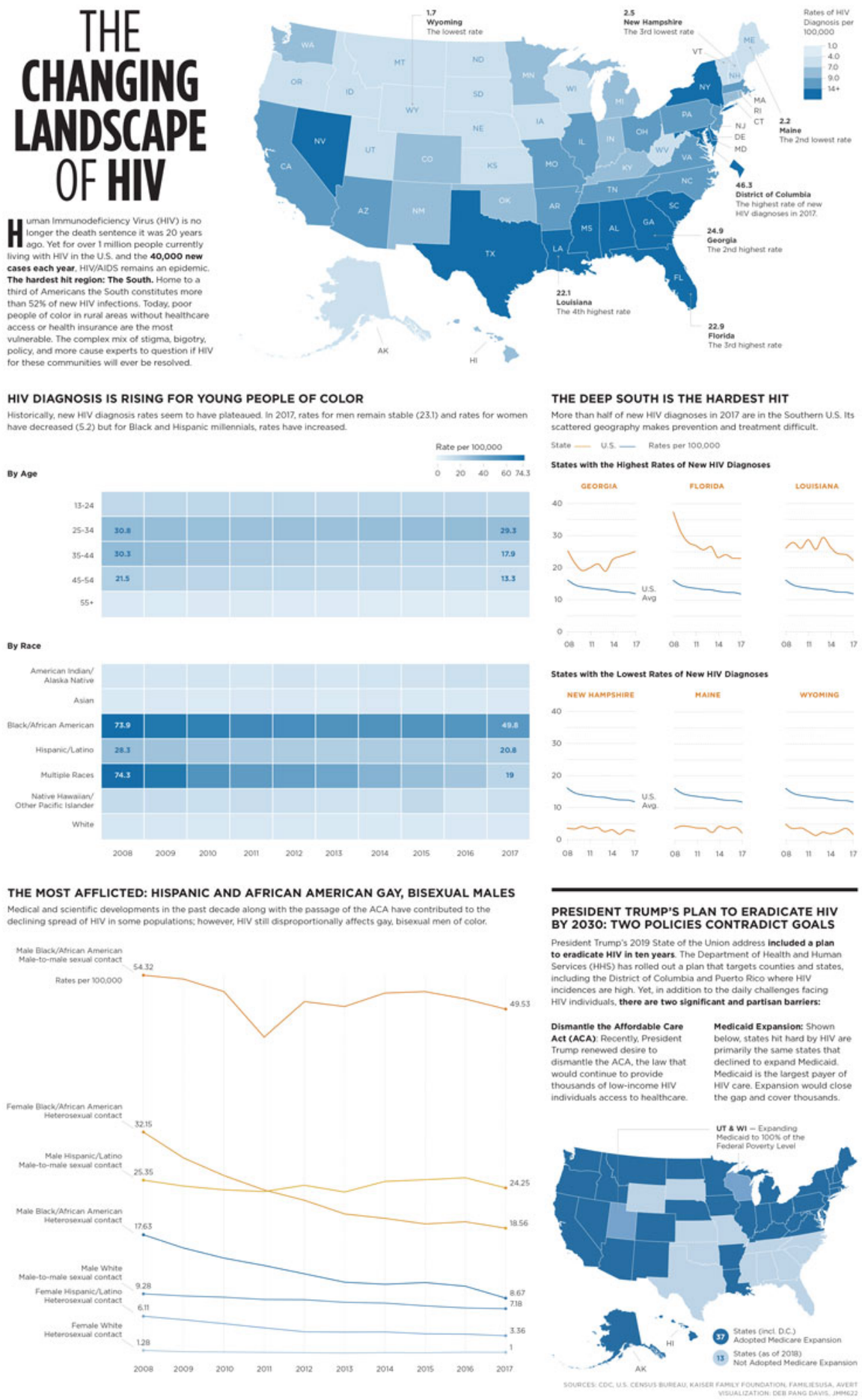
Women in the workforce

Female labor force participation rate, total %



Example by a former student

You can find these projects, and many others, in the Examples folder on Blackboard. Use them as inspiration for your work.



You can choose any topic for your infographic. If you don't know what topic to choose, visit https://www.gapminder.org/videos/ and see some lectures by Professor Hans Rosling.

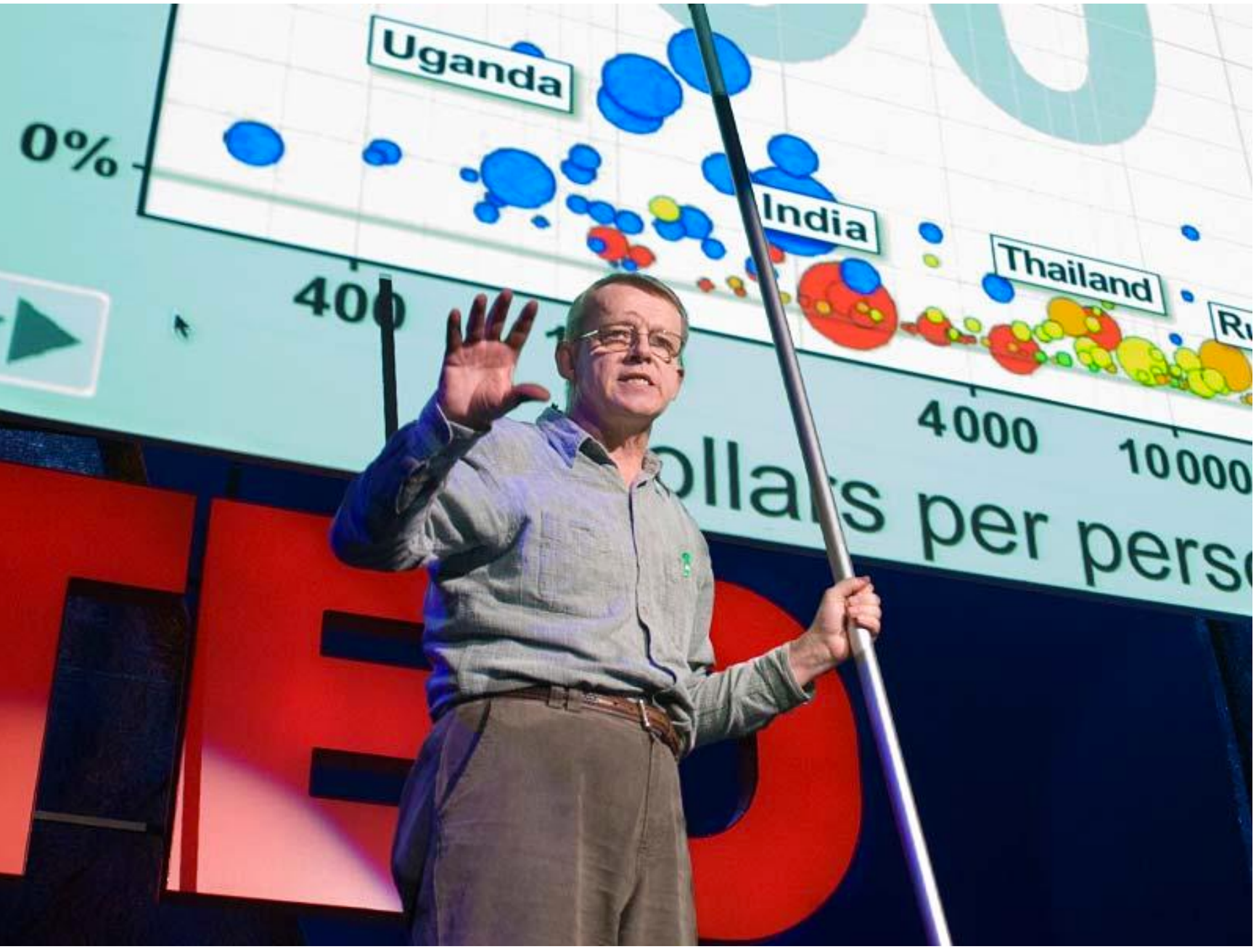
This is one of the most famous ones:
https://www.youtube.com/watch?v=jbkSRLYSojo&t=2s

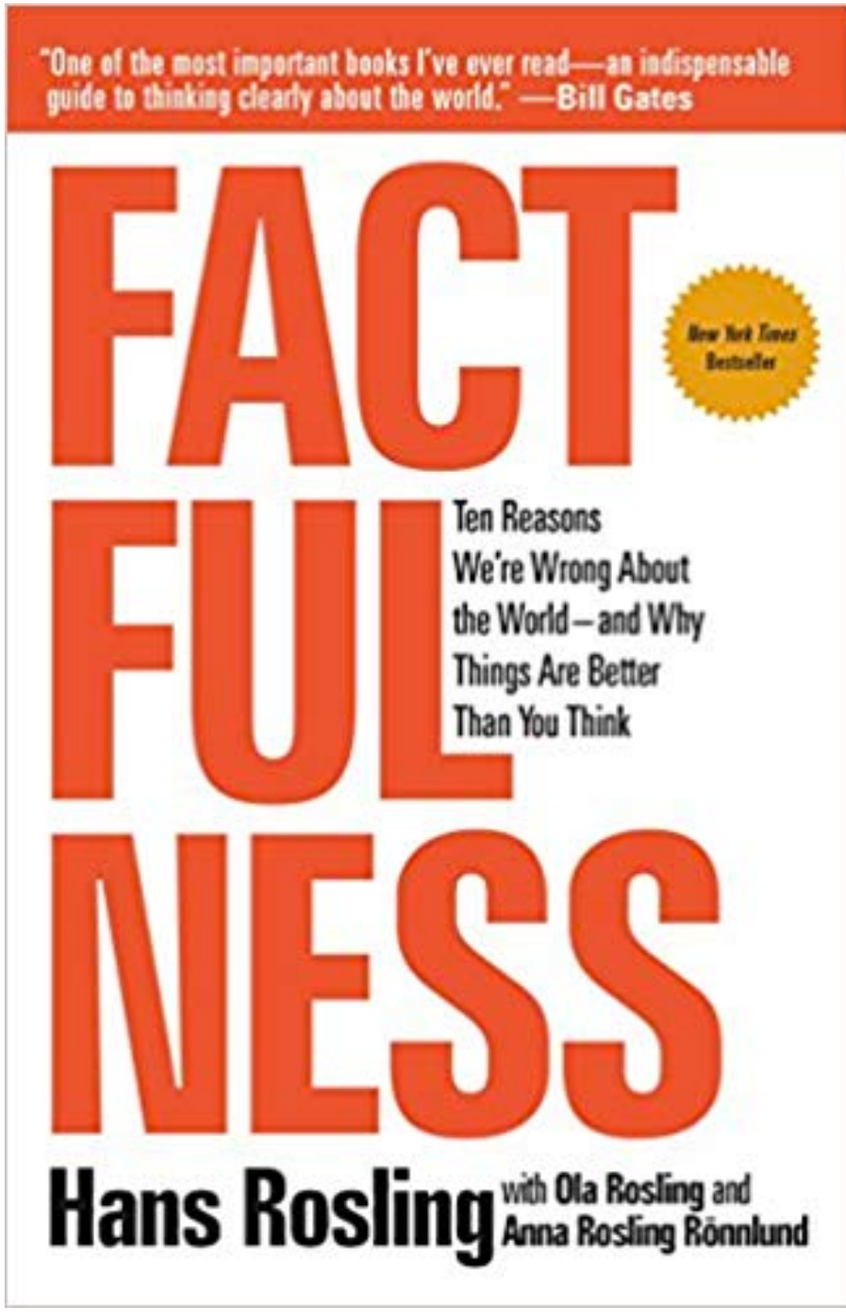
From this documentary:
https://www.youtube.com/watch?v=en2ix9f8ceM&list=PLBE30C2B39FE4BD1C

Next, imagine that you are asked to design a static infographic to explain some of the points that he made in his presentations.

To learn more about Professor Hans Rosling, see an interview with him in the last section of ***The Functional Art***

Hans Rosling's famous TED talk in 2006





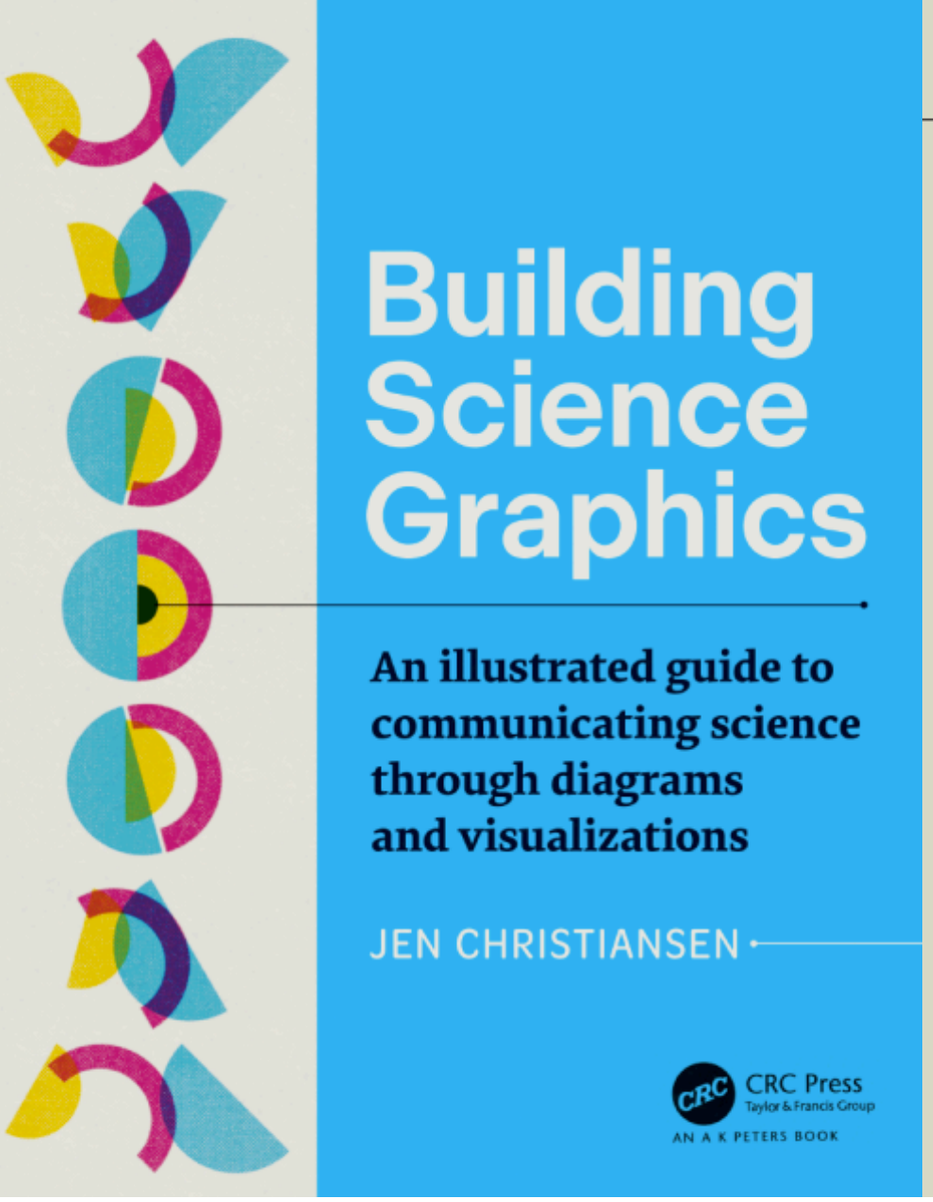
https://www.youtube.com/watch?v=hVimVzgtD6w

Where to find good data about worldwide social indicators

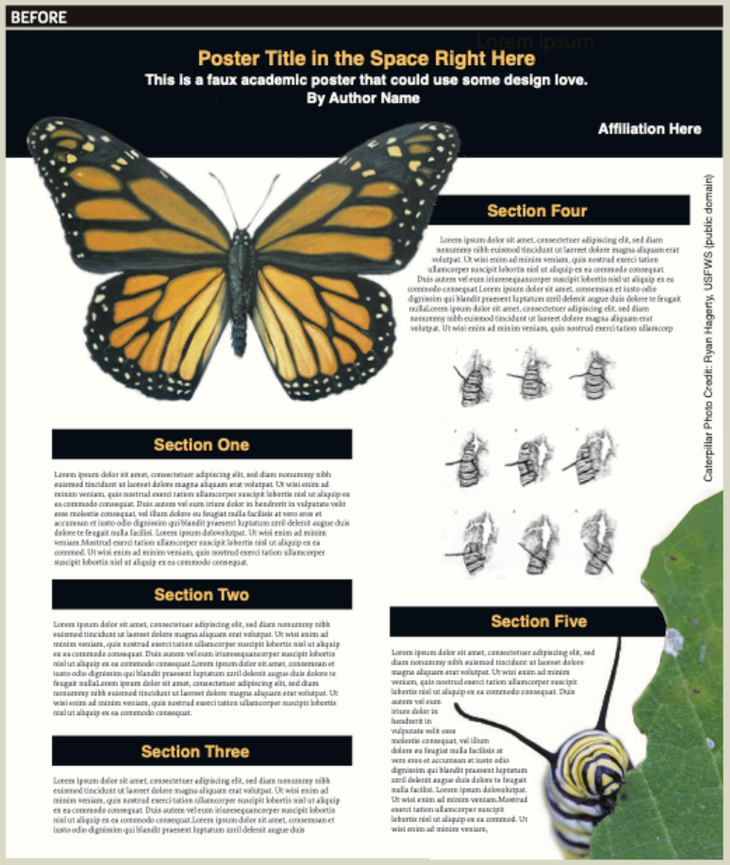
https://data.worldbank.org/

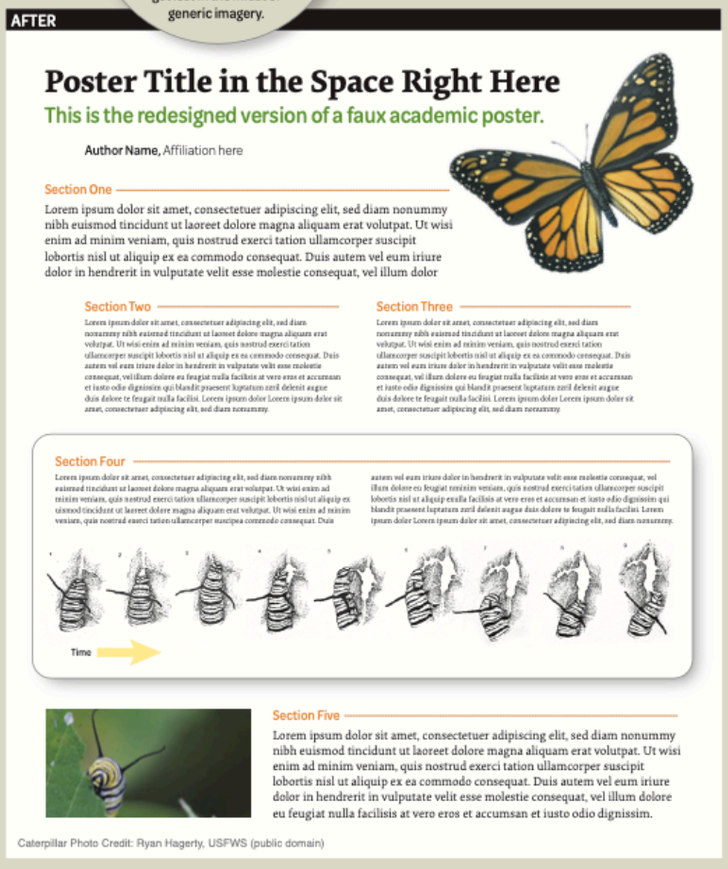
https://ourworldindata.org/

https://www.gapminder.org/data/



Engaging photos and drawings can help hook people and give them a quick sense of the subject matter of your poster. But I don't recommend crowding out the rest of your content with them. For academic conferences, make sure that the key take-home message of your research doesn't get lost in the midst of generic imagery.





Strongly recommended readings:

Chapter 7 of 'Building Science Graphics,' which you can find here:
https://www.dropbox.com/s/rpktdga0zc69cvg/BuildingScienceInfographics_Chapter7.pdf?dl=0

Chapter 8 of 'The Functional Art' (one of the textbooks on Blackboard)

An example of a visualization-driven story of mine inspired by Hans Rosling

Learn more about this project in:

The Functional Art

Chapter 8

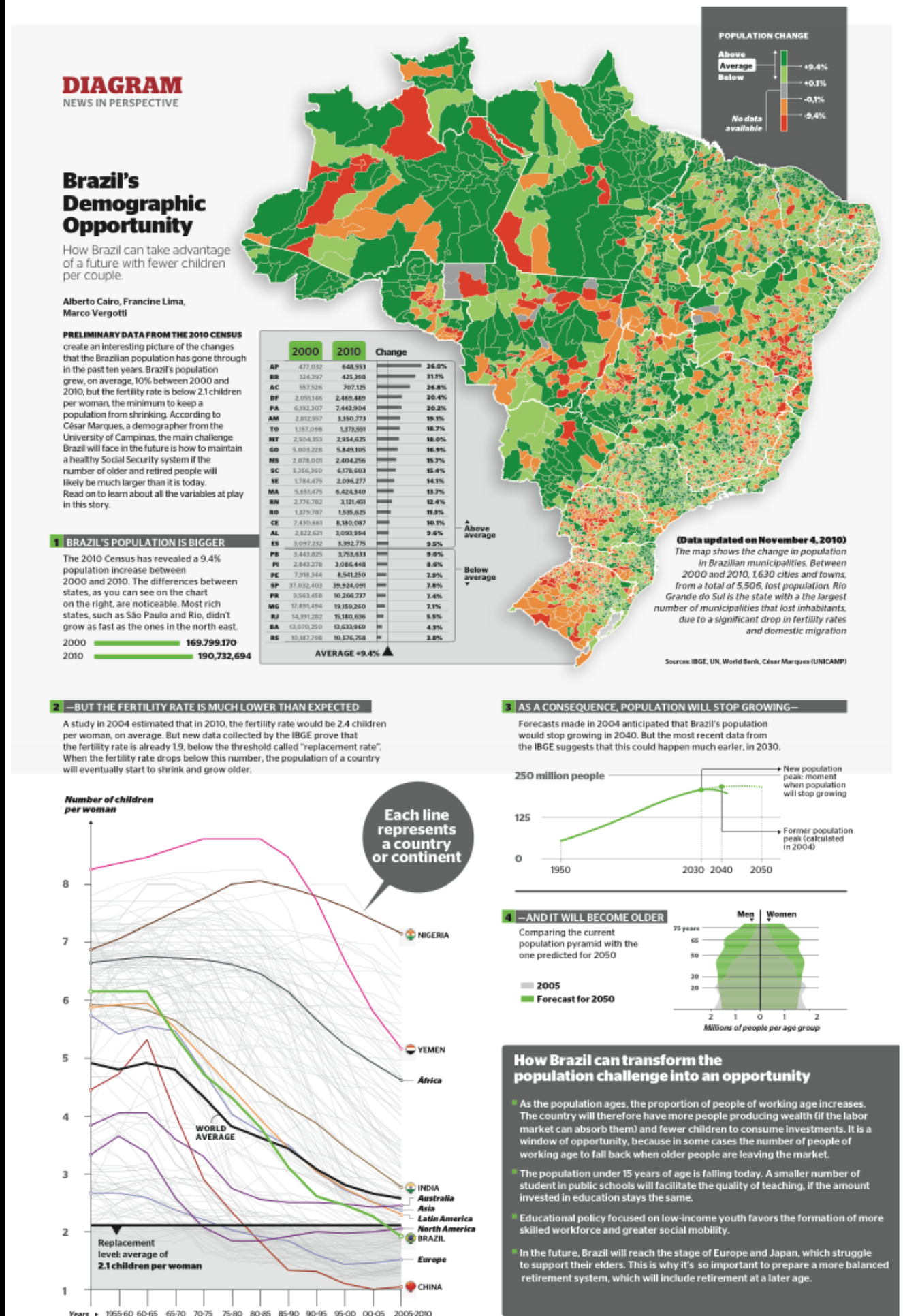


DIAGRAM
NEWS IN PERSPECTIVE

Brazil's Demographic Opportunity

How Brazil can take advantage of a future with fewer children per couple.

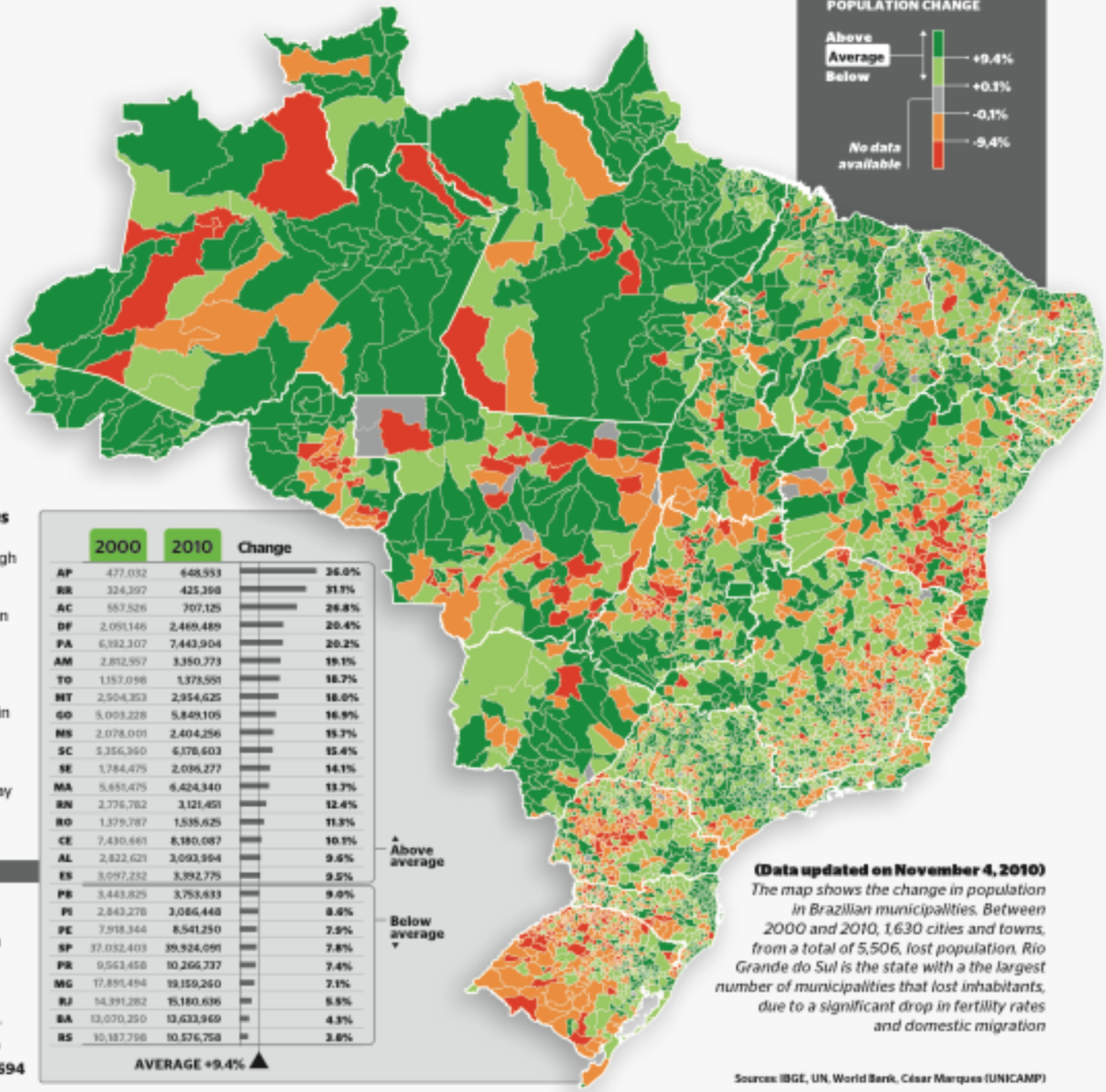
Alberto Cairo, Francine Lima, Marco Vergotti

PRELIMINARY DATA FROM THE 2010 CENSUS create an interesting picture of the changes that the Brazilian population has gone through in the past ten years. Brazil's population grew, on average, 10% between 2000 and 2010, but the fertility rate is below 2.1 children per woman, the minimum to keep a population from shrinking. According to César Marques, a demographer from the University of Campinas, the main challenge Brazil will face in the future is how to maintain a healthy Social Security system if the number of older and retired people will likely be much larger than it is today. Read on to learn about all the variables at play in this story.

1 BRAZIL'S POPULATION IS BIGGER

The 2010 Census has revealed a 9.4% population increase between 2000 and 2010. The differences between states, as you can see on the chart on the right, are noticeable. Most rich states, such as São Paulo and Rio, didn't grow as fast as the ones in the north east.

2000 **169.799.170**
2010 **190,732,694**



	2000	2010	Change
AP	477,032	648,553	36.0%
RR	324,397	425,398	31.1%
AC	557,526	707,125	26.8%
DF	2,051,146	2,469,489	20.4%
PA	6,192,307	7,443,904	20.2%
AM	2,812,557	3,350,773	19.1%
TO	1,157,098	1,373,551	18.7%
MT	2,504,353	2,954,625	18.0%
GO	5,003,228	5,849,105	16.9%
MS	2,078,001	2,404,256	15.7%
SC	5,356,360	6,178,603	15.4%
SE	1,784,475	2,036,277	14.1%
MA	5,651,475	6,424,340	13.7%
RN	2,776,782	3,121,451	12.4%
RO	1,379,787	1,535,625	11.3%
CE	7,430,661	8,180,087	10.1%
AL	2,822,621	3,093,994	9.6%
ES	3,097,232	3,392,775	9.5%
PB	3,443,825	3,753,633	9.0%
PI	2,843,278	3,086,448	8.6%
PE	7,918,344	8,541,250	7.9%
SP	37,032,403	39,924,091	7.8%
PR	9,563,458	10,266,737	7.4%
MG	17,891,494	19,159,260	7.1%
RJ	14,391,282	15,180,636	5.5%
BA	13,070,250	13,633,969	4.3%
RS	10,187,798	10,576,758	3.8%

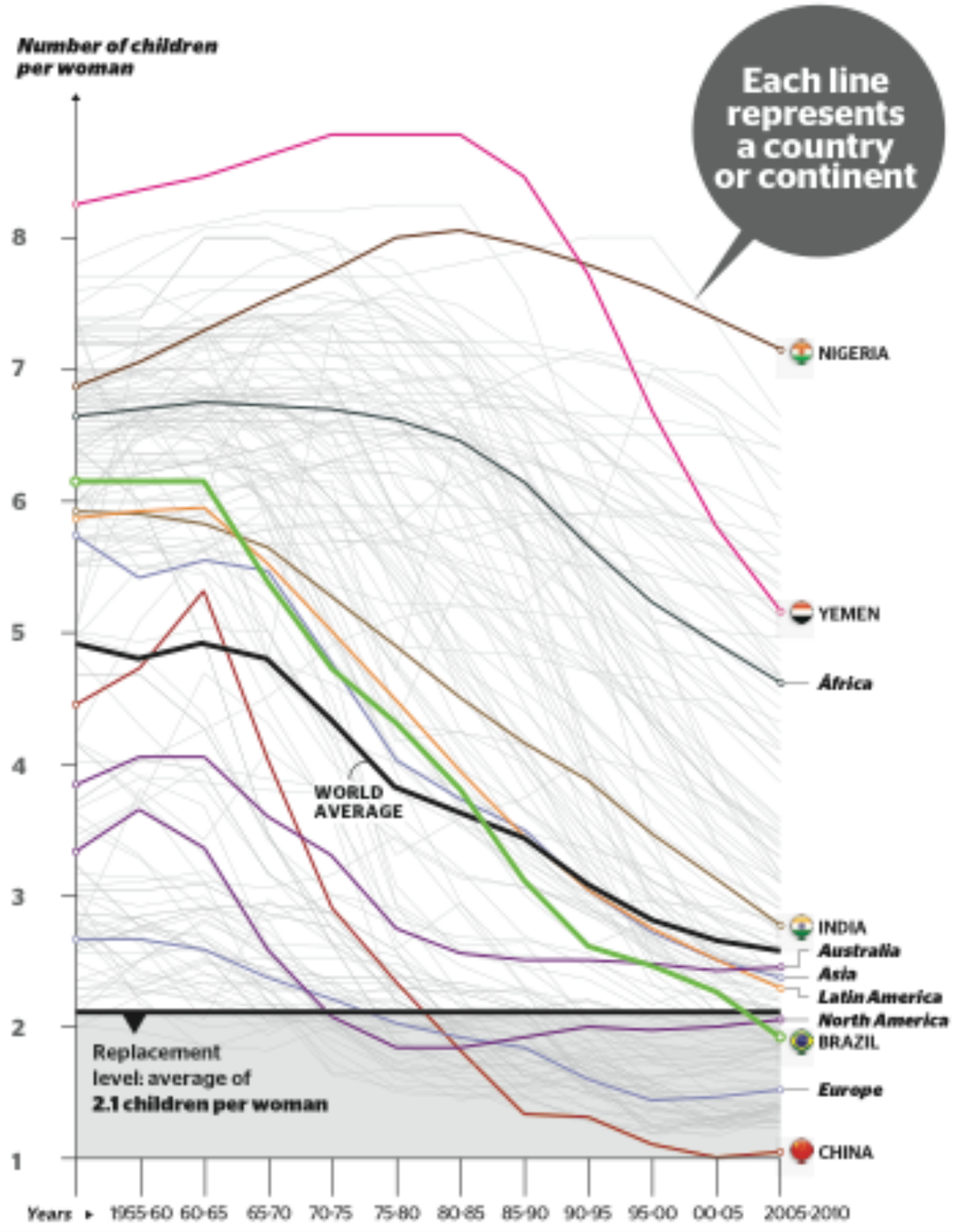
Above average
Below average
AVERAGE +9.4%

(Data updated on November 4, 2010)
The map shows the change in population in Brazilian municipalities. Between 2000 and 2010, 1,630 cities and towns, from a total of 5,506, lost population. Rio Grande do Sul is the state with a the largest number of municipalities that lost inhabitants, due to a significant drop in fertility rates and domestic migration

Sources: IBGE, UN, World Bank, César Marques (UNICAMP)

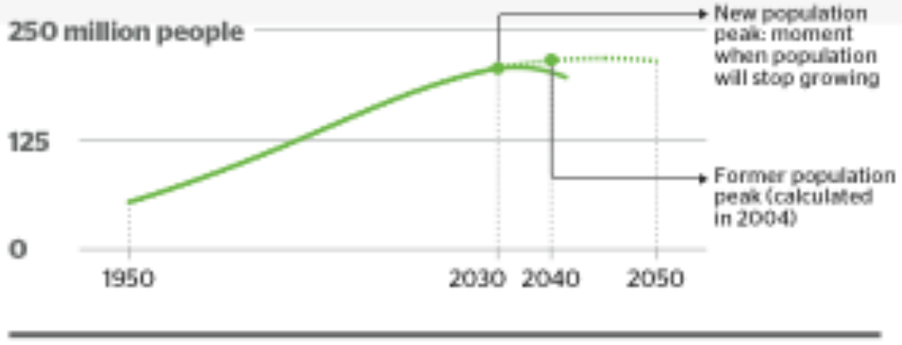
2 —BUT THE FERTILITY RATE IS MUCH LOWER THAN EXPECTED

A study in 2004 estimated that in 2010, the fertility rate would be 2.4 children per woman, on average. But new data collected by the IBGE prove that the fertility rate is already 1.9, below the threshold called "replacement rate". When the fertility rate drops below this number, the population of a country will eventually start to shrink and grow older.



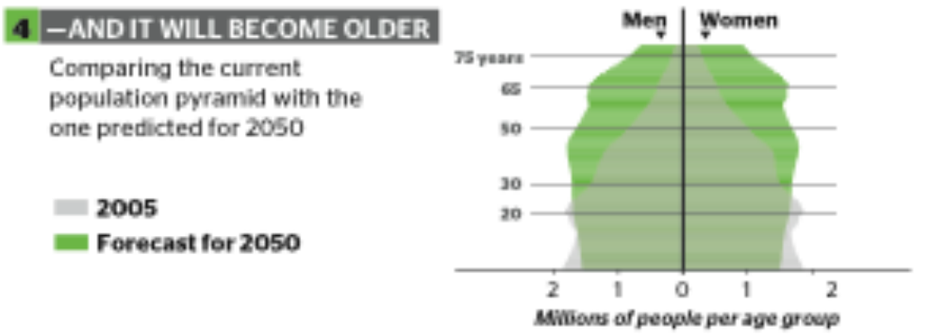
3 AS A CONSEQUENCE, POPULATION WILL STOP GROWING—

Forecasts made in 2004 anticipated that Brazil's population would stop growing in 2040. But the most recent data from the IBGE suggests that this could happen much earlier, in 2030.



4 —AND IT WILL BECOME OLDER

Comparing the current population pyramid with the one predicted for 2050



How Brazil can transform the population challenge into an opportunity

- As the population ages, the proportion of people of working age increases. The country will therefore have more people producing wealth (if the labor market can absorb them) and fewer children to consume investments. It is a window of opportunity, because in some cases the number of people of working age to fall back when older people are leaving the market.
- The population under 15 years of age is falling today. A smaller number of student in public schools will facilitate the quality of teaching, if the amount invested in education stays the same.
- Educational policy focused on low-income youth favors the formation of more skilled workforce and greater social mobility.
- In the future, Brazil will reach the stage of Europe and Japan, which struggle to support their elders. This is why it's so important to prepare a more balanced retirement system, which will include retirement at a later age.

How to structure a graphics story:

1. Begin by writing a long sentence that summarizes the points you want to make, and how they connect to each other

2. Split that long sentence into its components, using natural breaks

3. Transform those pieces into the sections of your infographic. Illustrate them with the appropriate charts, maps, diagrams, etc.

This project of mine is similar to the exercise you need to complete by September 15th (before the end of the day)